

Problem G. Toll Gates on a Tight Schedule

Input file: standard input
Output file: standard output
Time limit: 4 seconds
Memory limit: 1024 megabytes

You have to cross the city before the last train leaves.

The city has n junctions and m one-way roads. Each road takes exactly one minute to pass. Junction i has a toll gate, and passing through it costs c_i coins.

You start at junction 1 and want to reach junction n . Since you are in a hurry, you will only consider routes that use the minimum possible number of roads.

Among all such shortest routes, count how many have total toll at most C . The total toll of a route is the sum of c_i over all junctions on the route, including junctions 1 and n .

Because the answer may be large, output it modulo 2^{64} .

Input

The first line contains three integers n , m , and C ($1 \leq n \leq 90$, $0 \leq m \leq 300$, $0 \leq C \leq 10^{12}$).

The second line contains n integers $c_1, c_2, \dots, c_n$ ($0 \leq c_i \leq 10^{12}$), where c_i is the toll of junction i .

Each of the next m lines contains two integers u and v ($1 \leq u, v \leq n$, $u \neq v$), denoting a one-way road from u to v .

There are no repeated directed roads.

Output

Print one integer: the number of shortest routes from junction 1 to junction n whose total toll is at most C , modulo 2^{64} .

Example

standard input	standard output
4 5 8 1 2 3 4 1 2 1 3 2 4 3 4 2 3	2

Note

In the example, the shortest routes from junction 1 to junction 4 use exactly two roads.

They are $1 \rightarrow 2 \rightarrow 4$, with total toll $1 + 2 + 4 = 7$, and $1 \rightarrow 3 \rightarrow 4$, with total toll $1 + 3 + 4 = 8$.

Both routes have total toll at most 8.

The route $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$ is not counted, because it is not a shortest route.